THE NUMBER OF MINIMAL PRIMES OF A FILTERED COLIMIT OF FAITHFULLY FLAT MAPS IS THE SUPREMUM OF THE STAGES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 10302-10309 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 10311-10331, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Minimal primes of a ring, filtered colimits of rings, and faithful flatness; the comparison is retained.

Source reference. Original lemma algebra-lemma-colimit-faithfully-flat: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10331–10331. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma algebra-lemma-ff-rings: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10331–10331. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma algebra-lemma-flat-going-down: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10331–10331. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma algebra-lemma-minimal-prime-image-minimal-prime: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 10331–10331. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $R = \operatorname{colim} R_i$ be colimit of a directed system of rings whose transition maps are faithfully flat. Then the number of minimal primes of R taken as an element of $\{0, 1, 2, \ldots, \infty\}$ is the supremum of the numbers of minimal primes of the R_i.*

Date: stacks-project:more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. If $A \to B$ is a flat ring map, then $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ maps minimal primes to minimal primes by going down (Algebra, Lemma algebra-lemma-flat-going-down). If $A \to B$ is faithfully flat, then every minimal prime is the image of a minimal prime (by Algebra, Lemma algebra-lemma-ff-rings and algebra-lemma-minimal-prime-image-minimal-prime). Hence the number of minimal primes of R_i is $\geq$ the number of minimal primes of $R_{i'}$ if $i \leq i'$. By Algebra, Lemma algebra-lemma-colimit-faithfully-flat each of the maps $R_i \to R$ is faithfully flat and we also see that the number of minimal primes of R is $\geq$ the number of minimal primes of R_i. Finally, suppose that $\mathfrak{q}_1, \ldots, \mathfrak{q}_n$ are pairwise distinct minimal primes of R. Then we can find an i such that $R_i \cap \mathfrak{q}_1, \ldots, R_i \cap \mathfrak{q}_n$ are pairwise distinct (as sets and hence as prime ideals). This implies the lemma. $\square$